Empirical Evaluation of Federated Averaging and Federated Proximal Optimization under Non-IID Partitioning and Symmetric Label Noise on PathMNIST

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Abstract: Federated learning (FL) enables decentralized participants to collaboratively train a shared predictive model without pooling raw private observations. In digital histopathology applications, FL frequently confronts two simultaneous challenges: inter-client class distribution skew (non-IID data) and local annotation noise arising from subjective grading or tissue preparation artifacts. While Federated Proximal Optimization (FedProx) is hypothesized to mitigate local client drift by penalizing parameter deviation from the global model, empirical validation under paired non-IID conditions coupled with severe label noise on histopathological imaging remains limited.

In this work, we conduct a seed-controlled paired benchmark comparing Federated Averaging (**FedAvg**) against Federated Proximal Optimization (**FedProx**, $\mu = 0.01$) across five canonical random seeds (42, 123, 2026, 3407, 7777). We evaluate on PathMNIST, a public 9-class colorectal histopathology dataset (89,996 training, 10,004 validation, and 7,180 test images), partitioned into five simulated federated clients using a Dirichlet distribution ($\alpha = 0.3$). To assess resilience against annotation errors, Client 5 is subjected to exactly 40.00% symmetric label noise. Both algorithms were trained across 30 communication rounds using an identical 2-stage convolutional network (**CompactPathCNN**, 36,425 trainable parameters), identical seed-specific partitions, and identical initialization weights, totaling 300 executed communication rounds on an AMD Ryzen 5 5600H CPU execution platform.

Over the five seeds, FedAvg achieved a mean held-out test accuracy of 68.880% ± 9.266 pp (mean macro-F1: 0.6008 ± 0.0948), while FedProx achieved 68.279% ± 8.551 pp (mean macro-F1: 0.5934 ± 0.0960), yielding a paired difference of -0.602 ± 1.685 pp in test accuracy ($t(4) = -0.7983$, two-sided $p = 0.46938$, 95% CI: [-2.694 pp, +1.491 pp]) and -0.0075 ± 0.0105 in macro-F1 ($t(4) = -1.5908$, $p = 0.18686$, 95% CI: [-0.0205, +0.0056]). Under the evaluated configuration, **no statistically significant difference in test performance was detected between FedAvg and FedProx across the five random seeds**. Because five seeds provide limited statistical power and no formal equivalence test with predefined equivalence margins was conducted, this non-significant result does not establish statistical equivalence. Mechanistically, the proximal penalty penalizes parameter deviation from the global model but does not alter the corrupted gradient directions induced by noisy annotations. Consequently, under the evaluated configuration (PathMNIST, simulated five-client federation, Dirichlet $\alpha = 0.3$, single-client 40% symmetric noise, CompactPathCNN, and FedProx $\mu = 0.01$), proximal regularization alone did not overcome annotation noise without explicit noise filtering or adaptive client reweighting.

1. Introduction & Research Scope

Federated learning allows clinical and research centers to collaboratively train shared diagnostic models while keeping digitized histology images localized. However, decentralized histopathology introduces significant heterogeneity: (1) **Label Distribution Skew (Non-IID Data):** Specialized clinics observe disparate distributions of tissue pathologies. (2) **Annotation Noise (Symmetric Label Noise):** Staining variations, tissue folding, and inter-observer diagnostic disagreement generate mislabeled patches.

Operational Boundaries: This research utilizes the public PathMNIST dataset partitioned synthetically into five simulated client shards. No real-world multi-hospital network was deployed. Differential privacy was not implemented. Findings are bounded strictly to the evaluated configuration ($\mu = 0.01$, CompactPathCNN, PathMNIST, $\alpha = 0.3$, single-client 40% noise).

2. Experimental Setup & Methodology

Dataset: PathMNIST (MedMNIST v2 benchmark suite, 9 colorectal tissue classes, 28x28x3 RGB). Splits: 89,996 training, 10,004 validation, and 7,180 quarantined held-out test samples.

Model Architecture: CompactPathCNN is a lightweight 2-stage convolutional network with exactly **36,425 trainable parameters**. It comprises Conv1 (3→32, 3x3, ReLU, MaxPool), Conv2 (32→64, 3x3, ReLU, MaxPool), AdaptiveAvgPool2d(2, 2), Flatten (256-d), FC1 (256→64, ReLU), and FC2 (64→9 logits).

Partitioning & Noise Injection: 5 federated clients apportioned via Dirichlet distribution ($\alpha = 0.3$). Client 5 receives exactly 40.00% symmetric label noise modulo 9. Training ran for 30 communication rounds, 2 local epochs per round, batch size 32, learning rate 0.01 on 2 PyTorch CPU threads.

3. Five-Seed Experimental Benchmark Results

Seed	Dirichlet Client Counts	Client 5 Noisy Samples	FedAvg Test Acc (%)	FedProx Test Acc (%)	Paired Δ (pp)	FedAvg Macro-F1	FedProx Macro-F1	Δ Macro-F1	FedAvg Bal Acc (%)	FedProx Bal Acc (%)
42	24650, 30566, 14388, 12497, 7895	3,158 / 7,895 (40.0%)	79.624	79.638	**+0.014**	0.7403	0.7413	+0.0010	74.357	75.428
123	21522, 19760, 14721, 13041, 20952	8,381 / 20,952 (40.0%)	70.738	70.292	**-0.446**	0.6181	0.6136	-0.0045	64.794	64.316
2026	6502, 19450, 28146, 15764, 20134	8,054 / 20,134 (40.0%)	57.730	57.981	**+0.251**	0.5186	0.5133	-0.0053	56.294	56.307
3407	20279, 14263, 11886, 30337, 13231	5,292 / 13,231 (40.0%)	75.195	71.671	**-3.524**	0.6214	0.5957	-0.0257	65.631	63.183
7777	11406, 6714, 20804, 30392, 20680	8,272 / 20,680 (40.0%)	61.114	61.811	**+0.697**	0.5057	0.5030	-0.0027	54.802	55.278
Mean	—	—	**68.880**	**68.279**	**-0.602**	**0.6008**	**0.5934**	**-0.0075**	**63.176**	**62.902**
Std Dev (ddof=1)	—	—	*±9.266*	*±8.551*	*±1.685*	*±0.0948*	*±0.0960*	*±0.0105*	*±7.924*	*±8.072*

Statistical Hypothesis Testing (df = 4)

- **Test Accuracy:** Mean Paired Difference = -0.602 pp (±1.685 pp). Paired t(4) = -0.7983, p = 0.46938 (two-sided, fail to reject H_0). 95% Confidence Interval: [-2.694 pp, +1.491 pp]. Wilcoxon signed-rank W = 7.0, p = 1.000.
- **Test Macro-F1:** Mean Paired Difference = -0.0075 (±0.0105). Paired t(4) = -1.5908, p = 0.18686. 95% CI: [-0.0205, +0.0056]. Wilcoxon signed-rank W = 1.0, p = 0.125.
- **Best Validation Accuracy:** Mean Difference = -0.520 pp (±0.166 pp). Paired t(4) = -7.0096, p = 0.00218. 95% CI: [-0.726 pp, -0.314 pp]. Peak validation accuracy was slightly higher for FedAvg across all partitions (-0.520 ± 0.166 pp); we report this empirical difference directly without asserting an unsupported causal claim.
- **Test Loss:** FedAvg Mean = 0.9082 ± 0.1897; FedProx Mean = 0.9239 ± 0.1877 (Paired Δ = +0.0156 ± 0.0407, t(4) = 0.8566, p = 0.43994).

4. Scientific Figures

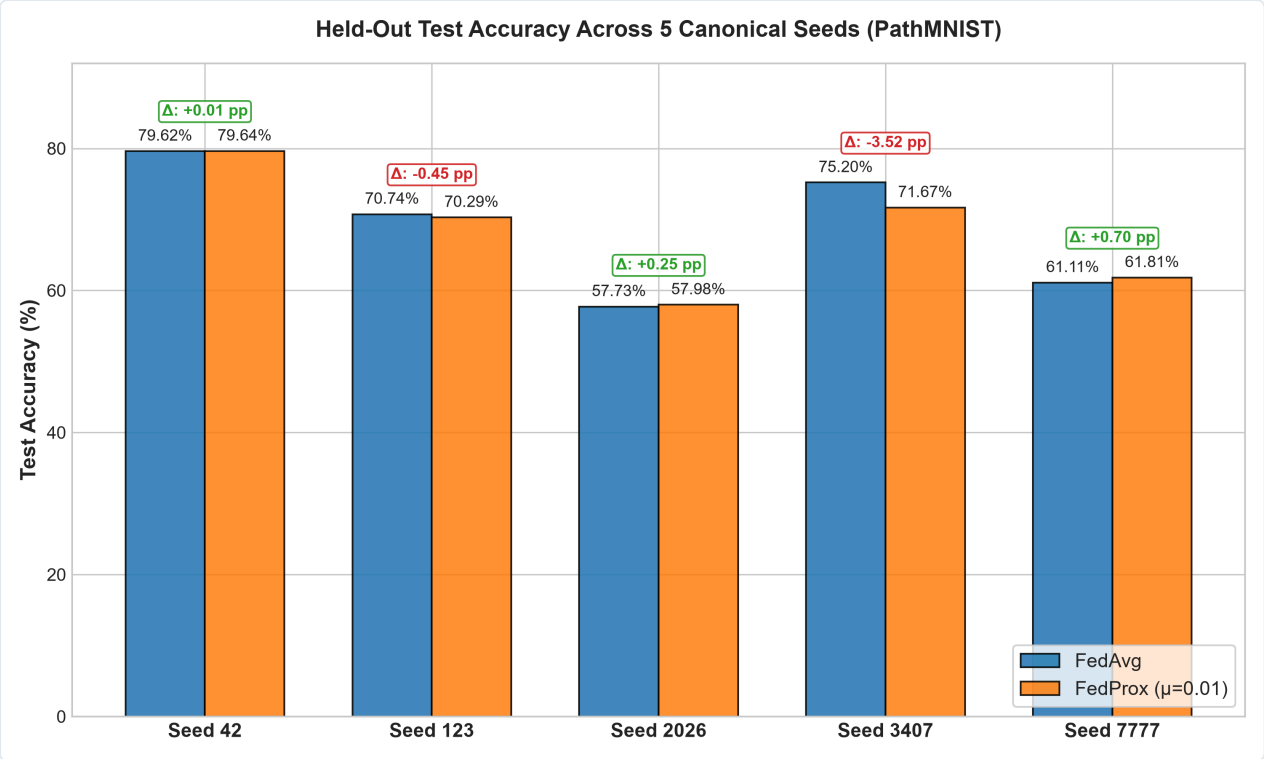


Figure 1: Test accuracy comparison across all five seeds (quarantined held-out test split).

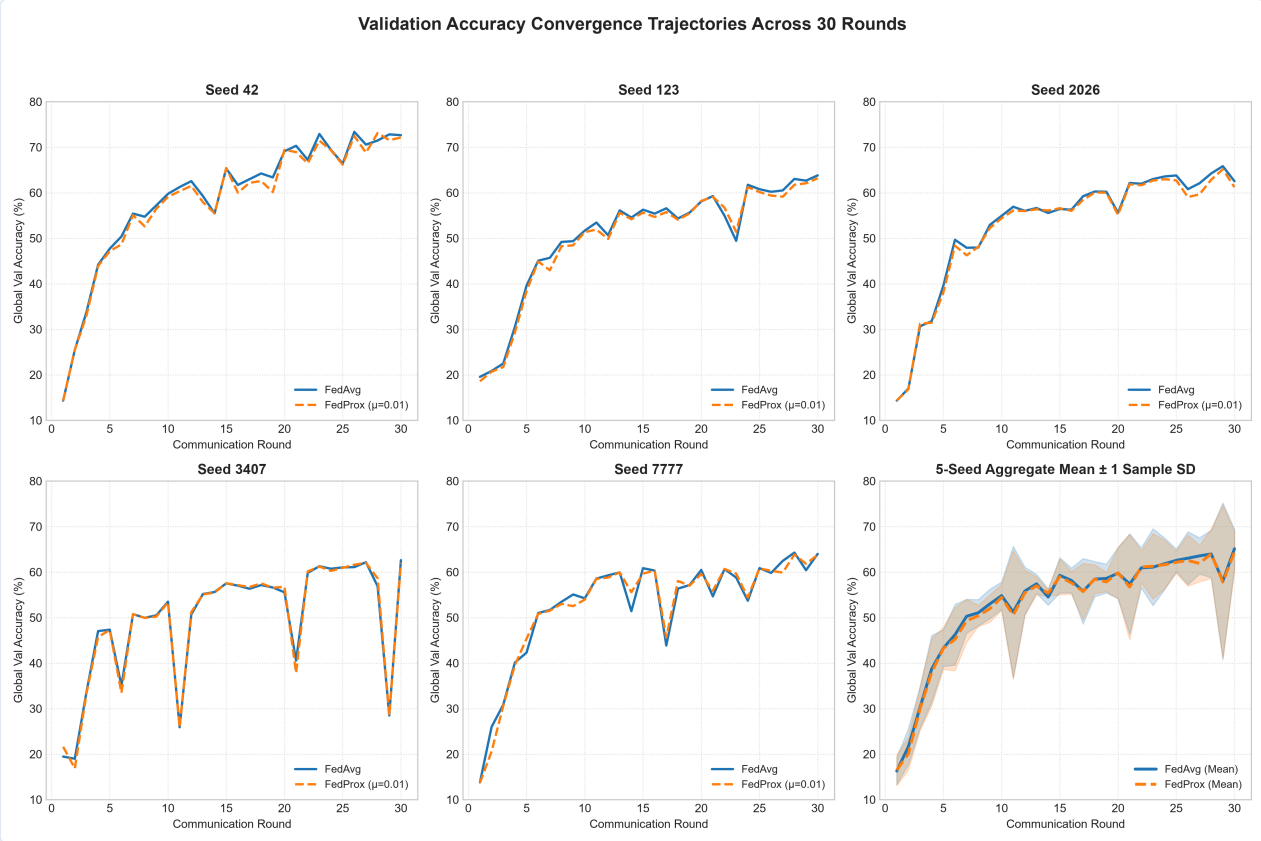


Figure 2: Validation convergence trajectories across 30 rounds for all five seeds and aggregate mean curve.

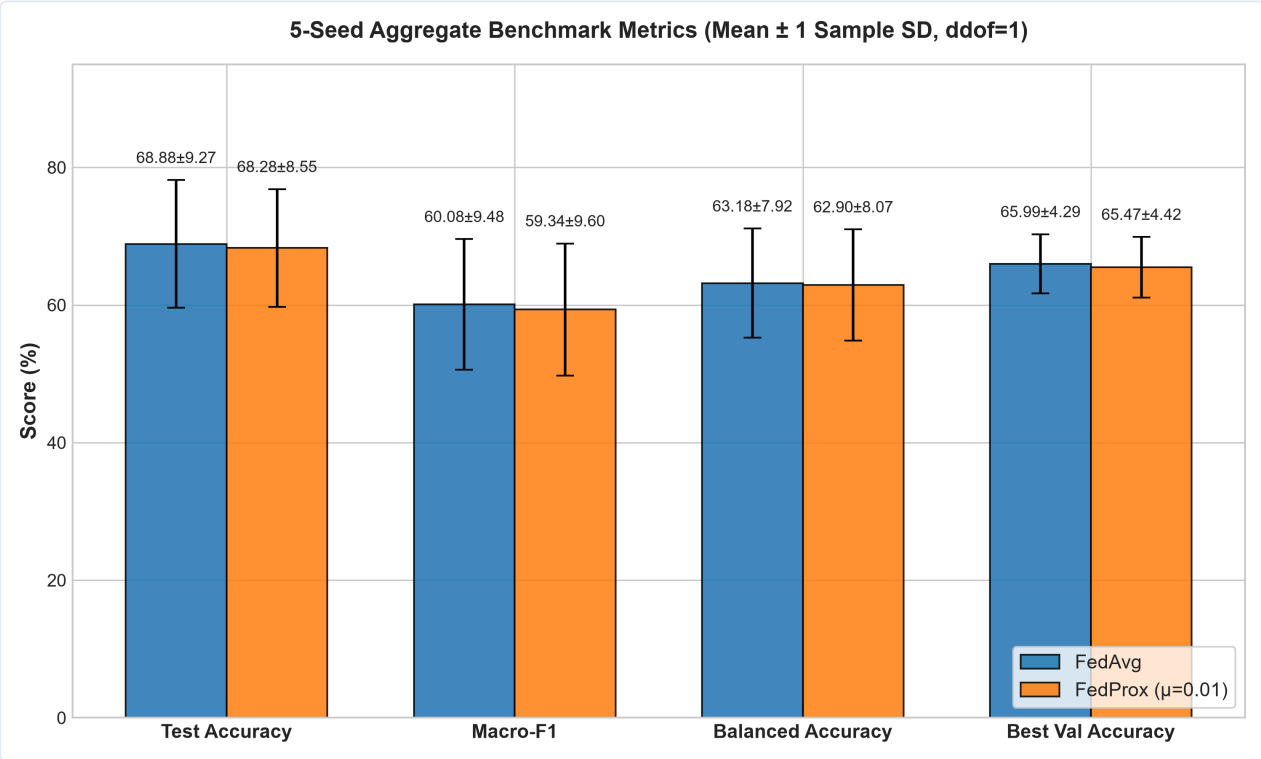


Figure 3: 5-seed aggregate benchmark metrics comparing FedAvg and FedProx (Mean ± 1 Sample Std Dev, ddof=1).



Figure 4: Per-class F1 score comparison across 9 histopathological tissue classes (averaged across 5 seeds).

5. Scientific Interpretation & Update Dynamics

Why Proximal Regularization Does Not Neutralize Label Corruption: FedProx's proximal term $\mu/2 \|w - w^t\|^2$ continuously penalizes parameter deviation from the global model. While this penalizes deviation magnitude, it *does not alter the corrupted gradient directions* induced by mislabeled samples. When aggregated via sample weighting, Client 5's contaminated model update still enters the global consensus proportionally to its sample count.

Update Cosine Similarities: Negative client-update cosine similarities were observed between Client 5 and clean clients. However, label noise cannot be claimed as the sole cause: Dirichlet label skew ($\alpha = 0.3$) creates severe client-level class imbalance, which independently produces orthogonal or divergent gradient vectors across clients. Both factors jointly contribute.

6. Hardware Telemetry & Runtime

The entire federated training pipeline ran on the CPU (AMD Ryzen 5 5600H, 2 threads). While CUDA diagnostics were confirmed during pre-flight, GPU compute was not used for training. Total compute across 300 rounds was 471.7 minutes (~7.86 hours): FedAvg averaged 42.86 ± 16.08 min/seed; FedProx averaged 51.48 ± 24.38 min/seed.

7. Conclusion & Recommendations

Under non-IID partitioning ($\alpha = 0.3$) and 40% single-client label noise on PathMNIST with CompactPathCNN and FedProx $\mu = 0.01$, **no statistically significant difference in test performance was detected between FedAvg and FedProx across five random seeds ($p = 0.469$)**. Because five seeds provide limited statistical power and no formal equivalence testing was conducted, this does not establish statistical equivalence. In this setting, proximal regularization alone did not overcome annotation noise, motivating explicit noise-aware defenses (e.g., Generalized Cross-Entropy loss or cosine similarity-based client reweighting).

References

1. McMahan, B., et al. (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data. *AISTATS*.
2. Li, T., et al. (2020). Federated Optimization in Heterogeneous Networks. *MLSys*.
3. Yang, J., et al. (2023). MedMNIST v2 - A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*.